

DESIGN CALCULATIONS

Transformer-Based Dual Regulated DC Power Supply (9V & 12V)

Design calculations reproduced from the supplied project report, including AC-to-DC conversion, ripple estimation, regulator input verification, capacitor selection, regulator power dissipation, diode stress, LED resistor selection, and transformer VA check.

4.1 Basic Relationships — From AC Secondary to DC

Secondary RMS voltage: 12 V (per half winding used in the bridge)

$$V_{rms} = 12 \text{ V}$$

Peak of AC waveform:

$$V_{peak} = V_{rms} \times \sqrt{2}$$

$$V_{peak} = 12 \times 1.414 \approx \mathbf{16.97 \text{ V}}$$

For the bridge rectifier, current flows through two silicon diodes. Assuming $V_f \approx 0.7 \text{ V}$ for each diode:

$$\text{Total diode drop} = 2 \times V_f = 2 \times 0.7 = \mathbf{1.4 \text{ V}}$$

Approximate DC peak after bridge and before filtering:

$$V_{DC, peak} \approx V_{peak} - 2V_f$$

$$V_{DC, peak} \approx 16.97 - 1.4 \approx \mathbf{15.57 \text{ V}}$$

Design note: the capacitor voltage rating should be above this peak. The report recommends $\geq 25 \text{ V}$ and notes 35 V as a safer option.

4.2 Ripple Frequency and Ripple Formula

With a full-wave bridge supplied from 50 Hz mains:

$$f_{ripple} = 2 \times f_{mains} = 2 \times 50 = \mathbf{100 \text{ Hz}}$$

For a capacitor-input filter, approximate ripple voltage is:

$$\Delta V_{pp} = I_{load} / (f_{ripple} \times C)$$

Where:

I_{load} = DC current drawn from the filtered bus

C = filter capacitance in farads

f_{ripple} = ripple frequency

Example 1 — $C = 2200 \mu\text{F}$ and $I_{load} = 0.5 \text{ A}$

$$\Delta V_{pp} = 0.5 / (100 \times 2200 \times 10^{-6})$$

$$\Delta V_{pp} \approx \mathbf{2.27 \text{ V}}$$

$$V_{min} \approx V_{DC, peak} - \Delta V_{pp}$$

$$V_{min} \approx 15.57 - 2.27 \approx \mathbf{13.30 \text{ V}}$$

Example 2 — $C = 4700 \mu\text{F}$ and $I_{load} = 0.5 \text{ A}$

$$\Delta V_{pp} = 0.5 / (100 \times 4700 \times 10^{-6})$$

$$\Delta V_{pp} \approx \mathbf{1.06 \text{ V}}$$

$$V_{\min} \approx 15.57 - 1.06 \approx \mathbf{14.51\text{ V}}$$

This keeps $V_{\min}$ above the assumed LM7812 dropout requirement of approximately 14 V for the 0.5 A example load.

4.3 Regulator Input Requirements and Verification

Assumed regulator headroom:

$$\text{LM7812: } V_{\text{in}}(\min) \approx 12 + 2 = \mathbf{14\text{ V}}$$

$$\text{LM7809: } V_{\text{in}}(\min) \approx 9 + 2 = \mathbf{11\text{ V}}$$

Using $V_{\min} \approx 14.51\text{ V}$ from the 4700 μF , 0.5 A example:

Regulator	Required $V_{\text{in}}(\min)$	Available $V_{\min}$	Verification
LM7812	$\approx 14\text{ V}$	$\approx 14.51\text{ V}$	$14.51\text{ V} \geq 14\text{ V}$ — OK
LM7809	$\approx 11\text{ V}$	$\approx 14.51\text{ V}$	$14.51\text{ V} \geq 11\text{ V}$ — OK

4.4 Filter Capacitor Value Selection

The practical capacitor-selection relation used in the report is:

$$C \geq I_{\max} / (\text{fripple} \times \Delta V_{\text{pp}, \max})$$

For $\Delta V_{\text{pp}, \max} = 1\text{ V}$, $I_{\max} = 1\text{ A}$ and $\text{fripple} = 100\text{ Hz}$:

$$C \geq 1 / (100 \times 1) = 0.01\text{ F} = \mathbf{10,000\text{ }\mu\text{F}}$$

Therefore, for a 1 A load and a 1 V ripple target, approximately 10,000 μF is required.

4.5 Regulator Power Dissipation and Heatsinking

Linear regulators dissipate the voltage drop as heat:

$$P_{\text{diss}} = (V_{\text{in}} - V_{\text{out}}) \times I_{\text{out}}$$

LM7812 at 1.0 A:

$$P = (15.57 - 12) \times 1.0 \approx \mathbf{3.57\text{ W}}$$

LM7809 at 1.0 A:

$$P = (15.57 - 9) \times 1.0 \approx \mathbf{6.57\text{ W}}$$

Design rule stated in the report: when dissipation is greater than approximately 1–2 W, a heatsink should be used.

4.6 Bridge Diode Stress and Selection

Approximate average diode power:

$$P_{\text{D}, \text{avg}} \approx V_{\text{f}} \times I_{\text{avg}}$$

For $I_{\text{avg}} = 1\text{ A}$ and $V_{\text{f}} = 0.7\text{ V}$:

$$P \approx 0.7 \times 1 = \mathbf{0.7\text{ W}}$$

Peak surge currents occur during capacitor-charging pulses, so the selected diodes must also be capable of handling the expected surge current.

4.7 LED Resistor Derivation

Assumptions used in the report: LED forward voltage $V_{\text{f}} \approx 1.5\text{ V}$ and desired LED current $\approx 10\text{ mA}$.

$$R = (V_{\text{reg}} - V_{\text{f}}) / I_{\text{LED}}$$

For the 12 V rail:

$$R = (12 - 1.5) / 0.01 = 1050 \, \Omega \approx \mathbf{1 \, k\Omega}$$

For the 9 V rail:

$$R = (9 - 1.5) / 0.01 = 750 \, \Omega$$

Using 1 k Ω gives approximately **7.5 mA** on the 9 V rail.

Resistor power on the 12 V rail:

$$P = (12 - 1.5) \times 0.01 \approx \mathbf{0.105 \, W}$$

Thus, a ¼ W resistor is sufficient for this example.

4.8 Transformer VA Check

$$\text{Transformer rating} = 12 \, \text{V} \times 1 \, \text{A} = \mathbf{12 \, VA}$$

The report specifies that the maximum continuous DC load should be $\leq 1 \, \text{A}$ to help prevent excessive heating and voltage drop.

Design Summary

Parameter	Calculated / Assumed Value
Transformer secondary	12-0-12 V, 1 A
Secondary RMS voltage	12 V
Peak secondary voltage	$\approx 16.97 \, \text{V}$
Bridge diode drop	$\approx 1.4 \, \text{V total}$
Filtered DC peak	$\approx 15.57 \, \text{V}$
Full-wave ripple frequency	100 Hz
LM7809 regulated output	9 V DC
LM7812 regulated output	12 V DC
Example capacitor	4700 μF gives $\approx 1.06 \, \text{V}$ ripple at 0.5 A
1 A / 1 V ripple target	$\approx 10,000 \, \mu\text{F}$
LM7812 dissipation at 1 A	$\approx 3.57 \, \text{W}$
LM7809 dissipation at 1 A	$\approx 6.57 \, \text{W}$
Transformer rating	12 VA

Source: Supplied project report, Chapter 4 — Design Calculations (pages 22–23).